Minimizing Latency for Controller Placement Problem in SDN

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***Abstract*—Software-defined networking (SDN) is a network technology that explicitly demarcates network control tasks from forwarding tasks. The infrastructure and the control plane are separated. It centrally specifies the control logic and manages the whole network. The network is controlled by a software program called a controller. A single controller is not sufficient to deal with the problem of scalability and dependability. To solve these problems, more number of controllers are required. Placing various controllers is called controller placement problem (CPP) in SDN. The problem is to select the count and location of controllers to be placed in the network. Control latency is an important metric in SDN as all functions are carried out through frequent message exchanges between the switches and controllers. Latency comprises mainly the processing, queuing, propagation and transmission latency. In a wide area network, propagation latency is an important metric which includes the average and worst case node to controller propagation latency. Besides, inter controller latency plays a significant role to maintain consistency in large networks. The sum total of switch-to-controller and inter-controller delay is known as the global latency. The objective of this paper is to improve SDN performance by lowering the network's overall latency in an optimal manner. A comparison of optimal with the random approach is shown in the results. Finding controller number while minimizing latency is another objective of this paper.**

***Index Terms*—Software defined network, controller placement problem, average and worst case latency, inter controller latency**

I. Introduction

The scale of networks has expanded dramatically as a result of the increasing number of devices that need connectivity, ranging from virtual machines to tablets and smart phones. The rapid rollout of applications designed using modern agile methods, which allows new apps to be released in weeks and days rather than months and years, is also driving the change of pace in system configuration. To make changes to these machines, a network administrator must connect to each one and make the necessary changes. To add to the difficulty, each system vendor has their own configuration language, user interface, and syntax, making it a time-consuming and highly cumbersome process. The traditional network become complex and sluggish and unmanageable because of the tight coupling of forwarding plane and control plane. A promising and new network paradigm which can address these problems is called the software defined network (SDN) architecture [1].

In SDN the control plane is centralized so the network configuration can be managed and updated in a highly agile and adaptable manner. As vendors strive to identify and incorporate SDN, it's become a grab word for a variety of networking technologies aimed at making networks more dynamic. By allowing the network to be programmed directly, which is possible due to the decoupling of the infrastructure and control plane, benefits such as boosting network utilisation efficiency, enabling network innovation, simplifying network management and so on can be realised.

Fig. 1 depicts the software defined network architecture. It is made up of three layers: the bottom infrastructure layer, the middle control layer and a top application layer. Controllers make the decision for forwarding traffic which takes place in the infrastructure layer also known as the data plane. Controller is a separate entity which is responsible for routing and signaling. For the forwarding devices such as switches and routers, rules and policies are generated with the help of controllers. Network traffic is handled by the control plane. The term management plane is used instead of application

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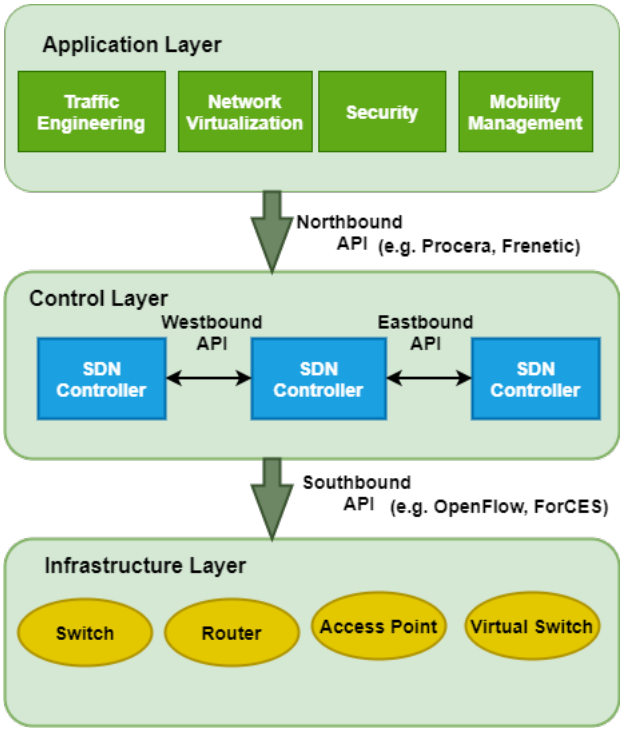

Fig. 1. SDN Architecture

plane by some researchers. The management plane is in charge of establishing rules and regulations. Besides these layers, three application programming interfaces are present to communicate between these layers. The southbound interface such as OpenFlow is responsible for communication between forwarding devices and the controllers [2] [3]. The northbound interface allows controllers and applications to communicate with one another. In a multi-controller scenario, the east-west bound API manage communication between groups of controllers.

An important research issue in SDN is the controller placement problem (CPP). The challenge of CPP, which Heller et. al [4] initially suggested in 2012, is calculating the count and deployment position of required controllers. The following questions arise during the installation of controllers in SDN [5]:

- What is the bare minimum number of controllers required for a network?
- Where should the controller(s) be deployed?
- How many networking devices can be linked to a single controller?

While addressing CPP problem, latency is an important performance measure. Due to the considerable influence of latency on network performance, network latency minimization is a key objective for the planning of networks. Consequently, literature provides frequent strategies to reduce the delay between switches and controllers [4] [6] [7]. Inter controller latency is crucial for many controllers in SDN, along with switch to controller latency, as the consistency of several controllers is substantially affected [8]. Most of the past research has attempted to tackle the CPP problem by lowering switch to controller delay while neglecting inter controller latency and network overall latency. The sum total of switch to controller and inter controller delay is the overall or global latency. The aim of this paper is to improve SDN performance by lowering the network's overall latency.
The major contributions are summarized below:

- The goal is to find the best possible location for k controllers while lowering the overall worst case and average propagation latency which includes the switch controller and inter controller latency considering controller's capacity.
- Comparison of optimal with the random approach.
- Find out optimal controller number while minimizing latency.
- Three networks from topology zoo are considered for evaluation.

The background and associated work are presented in the next section. problem statement is elaborated in section III. The evaluation on real networks are done and described in section IV. Conclusions and some future works are discussed at last.

II. Background and Related Work

Due to the frequent exchange of messages between switches and controllers, the latency is an important measure for all SDN functions. There are four types of control latency. [1] [5]

1. **Processing Latency.** This is affected by the service rate of controller.
2. **Queuing Latency.** The controller site's queuing latency is determined by its load and service rate.
3. **Propagation Latency.** The distance between switches and controllers or the path length determines the propagation latency.
4. **Transmission Latency.** The port rates at the switch and controller determines the transmission latency.

For different networks, the weight of the above mentioned time component varies. The controller processing latency and packet propagation time are predominant, while the delay for queuing of switches and packet transmission time are not very significant for large scale network spanning across geographically dispersed range. Therefore, the focus of research is to reduce the network latency in SDN for wide area networks considering controller processing delay and packet propagation delay. In this paper, propagation latency is considered as a performance measure.

The amount of time required for packets to move between two network nodes (switch and controller) is called the

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network latency. Latency is described as a distance between two nodes, which can be either a unit weight distance (also known as one hop distance) or a weighted distance. Delay in CPP might be inter-controller, switch-to-controller, switch-to-switch, or overall network latency. Most of the research work has attempted to tackle the CPP problem by lowering switch to controller delay while neglecting inter controller latency and total network latency.

When multiple controllers are needed to handle a network, the communication overhead between controller increases, while the network's overall performance improves. Switches can also manage a limited number of outgoing and incoming requests (receiving and sending packets). The packets may be dropped if the switch is overloaded. The sum of switch to controller and inter controller delay is known as the global latency. The goal of the research is to reduce the network's overall latency.

Most of the authors consider node to controller delay as a performance measure for controller placement optimization as illustrated by Heller et al. [4], Wang et. al. [6] and Ahmadi et. al. [7]. Heller considered both average and worst case latency as the performance criteria. Some other authors consider capacitated CPP considering the load of the switches while addressing CPP as Sallahi et al. [9], Mohanty et al. [10] and Yao et al [11]. Sallahi considered cost as a performance metric which incorporate the cost for deploying controllers, link cost of connecting switches with controllers and inter controller link costs. A dynamic controller provisioning is addressed by Bari et al. using simulated annealing [12]. He et al. also taken flow setup time in dynamic traffic scenario [13]. Chai et al. [14] considered switch-controller association while solving capacitated CPP. Some of the authors considered both switch-to-controller and inter -controller delay while solving CPP [8] [15] [16]. Very few authors considered the overall global latency while placing controllers [14] [15] [17].

III. Problem Formulation

The network is depicted as a graph $G = (V, L)$, where $V = Z \cup X$ are the nodes. $Z = \{z_1, z_2, ..., z_n\}$ be the set of switches, $X = \{x_1, x_2, ..., x_k\}$ be the set of controllers and $L = \{l_1, l_2, ..., l_e\}$ be the link set connecting these nodes.
Let $K = 2^V - 1$ denotes the set of all possible placement of controllers. Let $D(z, x)$ be the minimum distance or shortest communication path from switch z_i to controller x_j and $D(x_i, x_j)$ be the shortest inter controller communication link between x_i to x_j. The load of switch z_i is ρ_z and μ_x be the the capacity of controller x_j. A binary variable B_{zx} is set to 1 if switch z_i is allocated to controller x_j, otherwise it is set as 0.

TABLE I
Notations

Variables	Description
$G(V, L)$	Network graph
$V = Z \cup X$	Network nodes
L	Network links
Z	Forwarding devices
X	Set of controllers
K	Placement or positions for deploying controllers
$D(z, x)$	Shortest path latency between nodes $z, x \in V$
$D(x_i, x_j)$	Shortest path inter controller latency $x_i, x_j \in X$
ρ_z	Load of switch
μ_x	controller capacity
A_x	Boolean variable, set to 1 when a controller is located at x_j
B_{zx}	Boolean variable, set to 1 when switch is allocated to controller

$$\lambda_{sc-avgL}(K) = \frac{1}{|V|} \sum_{z \in Z} \min_{x \in K} D(z, x) B_{zx} \tag{1}$$

$$\lambda_{sc-worstL}(K) = \max_{z \in Z} \min_{x \in K} D(z, x) A_x \tag{2}$$

To maintain network state consistency and availability, inter controller latency is an important metric. The inter controller latency for average case $\lambda_{cc-avgL}(K)$ and worst case $\lambda_{cc-worstL}(K)$ are calculated as,

$$\lambda_{cc-avgL}(K) = \frac{1}{|K|} \sum_{x_i, x_j \in K} D(x_i, x_j) \tag{3}$$

$$\lambda_{cc-worstL}(K) = \max_{x_i, x_j \in K} D(x_i, x_j) \tag{4}$$

Total average and worst case latency are calculated as,

$$\mathcal{T}_{avgL}(K) = \lambda_{sc-avgL}(K) + \lambda_{cc-avgL}(K) \tag{5}$$

$$\mathcal{T}_{worstL}(K) = \lambda_{sc-worstL}(K) + \lambda_{cc-worstL}(K) \tag{6}$$

The goal is to minimize the global latency i.e., total average latency $\mathcal{T}_{avgL}(K)$ and total worst case latency $\mathcal{T}_{worstL}(K)$ which includes the switch to controller and inter controller latency subject to some constraints (9)-(12).

$$\text{minimize} \quad \mathcal{T}_{avgL}(K) \tag{7}$$

$$\text{minimize} \quad \mathcal{T}_{worstL}(K) \tag{8}$$

subject to,

$$\sum_{z \in Z} \rho_z B_{zx} \leq \mu_x A_x \quad \forall x \in K \tag{9}$$

$$\sum_{x \in K} B_{zx} = 1 \quad \forall z \in Z \tag{10}$$

$$B_{zx} \leq A_x \qquad \forall z \in Z, \forall x \in K \tag{11}$$

$$A_x, B_{zx} \in \{0, 1\}, \quad \forall z \in Z, \forall x \in K \tag{12}$$

The sum of switches load should not exceed the controller capacity as captured by(9). A single controller is allocated to each switch as in (10) and constraint(11) states that a switch is allocated to a controller if it is active. The binary variables are described in (12).

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IV. Result and Analysis

MATLAB 2020 was used for writing the programs and was executed on a machine with windows 10 Pro (64 bit) and 48 GB RAM. The data for the evaluation is taken from topology zoo [18], which is in graphical format. It is a collection of diverse datasets. Haversine formula is used to calculate the shortest distance linking two points on the Earth's surface. We applied the Bellman ford technique to find the shortest path between two nodes. All conceivable combinations of switch- to-controller and intercontroller latency are considered to figure out the appropriate controller combination in CPP. Three network datasets of varying sizes are taken from the topology zoo for evaluation: NSFNET (13 nodes and 18 edges), Agis (25 nodes and 30 edges) and Geant (37 nodes and 58 edges).

A. Performance based on Latency

Fig. 2 and 3 shows the result of average and worst case latency against the number of controllers in NSFNET, Agis and Geant network topology. The total or global latency is the sum of the latency between the switch and controller as well as the latency between controllers. The switch to controller latency is decreasing along with increasing controller number while inter controller latency is increasing for both average and worst case. With the growing number of controllers, global latency is lowering in Agis network but remains almost same in NSFNET and Geant networks as shown in fig. 2 and table II. In fig. 3 and table II, global latency is decreasing along with the increase in number of controllers for Agis network and is increasing for NSFNET and Geant networks.

Fig. 4 and 5 shows the comparison of optimal approach with random approach for both average and worst case latency. From these figures it can be shown that the optimal approach is better compared to the random approach.

B. Execution Time

Table II shows the execution time against controller number and global latency. The execution time slightly increases with increasing controller number for NSFNET. But in Agis and Geant the value is significantly very high. This is due to bigger network size of Agis and Geant compared to NSFNET.

C. Optimal Number of Controllers

Switches are allocated to controllers on the basis of controller capacity. After analyzing the results and the distribution of nodes inside the network infrastructure, the appropriate number of controllers required for a given network are evaluated. From the analysis it can be clerly seen that in NSFNET with 13 nodes one controller is sufficient. For Agis and Geant two to three controllers are required to manage its structure.

V. Conclusions

The network operators of SDN has significant hurdles in determining the appropriate number of controllers required and their positions while designing its architecture. To improve performance, the controller placement problem defines the number of controllers and their placements.The aim of this paper is to reduce the global latency in an optimal manner. The overall switch-to-controller and intercontroller delay is included in the global latency. Three network datasets are used to verify the results. The optimal approach is compared with the random approach showing the effectiveness of the optimal method. Another goal is to determine the number of controllers required to maintain the network structure. In the future scope, other performance metrics will be applied to a variety of real world architecture like cost, dependability and load balancing.

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References

[1] T. Das, V. Sridharan and M. Gurusamy, "A Survey on Controller Placement in SDN," in IEEE Communications Surveys & Tutorials, vol. 22, no. 1, pp. 472-503, Firstquarter 2020, doi: 10.1109/COMST.2019.2935453.

[2] Abha Kumari and Ashok Singh Sairam, "A Survey of Controller Placement Problem in Software Defined Networks," arXiv preprint arXiv:1905.04649 (2019).

[3] K. S. Sahoo, S. Mohanty, M. Tiwary, B. K. Mishra, and B. Sahoo, "A comprehensive tutorial on software defined network: The driving force for the future internet technology," in Proceedings of the International Conference on Advances in Information Communication Technology & Computing. ACM, 2016, p. 114.

[4] B. Heller, R. Sherwood and N. McKeown, "The controller placement problem," In Proceedings of the first workshop on Hot topics in software defined networks (HotSDN '12), NY, USA, 7–12. DOI:https://doi.org/10.1145/2342441.2342444

[5] G. Wang, Y. Zhao, J. Huang and W. Wang, "The Controller Placement Problem in Software Defined Networking: A Survey," in IEEE Network, vol. 31, no. 5, pp. 21-27, 2017, doi: 10.1109/MNET.2017.1600182.

[6] G. Wang, Y. Zhao, J. Huang, and Y. Wu, "An effective approach to controller placement in software defined wide area networks," IEEE Trans. Netw. Service Manag., vol. 15, no. 1, pp. 344–355, Mar. 2018.

[7] V. Ahmadi and M. Khorramizadeh, "An adaptive heuristic for multiobjective controller placement in software-defined networks," Comput. Elect. Eng., vol. 66, pp. 204–228, Feb. 2018. http://www.sciencedirect.com/science/article/pii/S0045790617307565

[8] S. Lange, S. Gebert, T. Zinner, P. Tran-Gia, D. Hock, M. Jarschel, M. Hoffmann, "Heuristic Approaches to the Controller Placement Problem in Large Scale SDN Networks," in IEEE Transactions on Network and Service Management, vol. 12, no. 1, pp. 4-17, March 2015, doi: 10.1109/TNSM.2015.2402432.

[9] A. Sallahi and M. St-Hilaire, "Optimal Model for the Controller Placement Problem in Software Defined Networks," in IEEE Communications Letters, vol. 19, no. 1, pp. 30-33, Jan. 2015, doi: 10.1109/LCOMM.2014.2371014.

[10] S. Mohanty, P. Priyadarshini, S. Sahoo, B. Sahoo and S. Sethi, "Metaheuristic Techniques for Controller Placement in Software-Defined Networks," TENCON 2019 - 2019 IEEE Region 10 Conference (TENCON), 2019, pp. 897-902, doi: 10.1109/TENCON.2019.8929265.

[11] G. Yao, J. Bi, Y. Li, and L. Guo, "On the capacitated controller placement problem in software defined networks," IEEE Commun. Lett., vol. 18, no. 8, pp. 1339–1342, Aug. 2014.

[12] M. F. Bari et al., "Dynamic Controller Provisioning in Software Defined Networks," Proceedings of the 9th International Conference on Network and Service Management (CNSM 2013), 2013, pp. 18-25, doi: 10.1109/CNSM.2013.6727805.

[13] M. He, A. Basta, A. Blenk and W. Kellerer, "Modeling flow setup time for controller placement in SDN: Evaluation for dynamic flows," 2017 IEEE International Conference on Communications (ICC), 2017, pp. 1-7, doi: 10.1109/ICC.2017.7996654.

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TABLE II
NETWORK TOPOLOGY (TOTAL AVERAGE AND WORST CASE LATENCY) AND EXECUTION TIME

Topology	K	$\mathcal{T}_{avgL}(K)$	Execution Time (in ms)	K	$\mathcal{T}_{worstL}(K)$	Execution Time (in ms)
NSFNET	2	6.5462	0.1257	2	12.241	0.1401
	3	6.4322	0.3554	3	16.363	0.3752
	4	6.2197	0.8518	4	18.605	0.9338
	5	6.2065	1.5912	5	24.253	1.6838
Agis	2	6.651	0.3831	2	19.62	0.5623
	3	4.5082	2.656	3	14.527	2.9811
	4	3.9803	15.653	4	12.249	15.502
	5	3.5473	72.748	5	14.694	69.29
Geant	2	5.2327	0.7786	2	13.316	3.5289
	3	5.0588	9.4324	3	14.846	22.839
	4	5.1408	87.475	4	15.88	89.623
	5	4.8697	602.34	5	15.744	607.04

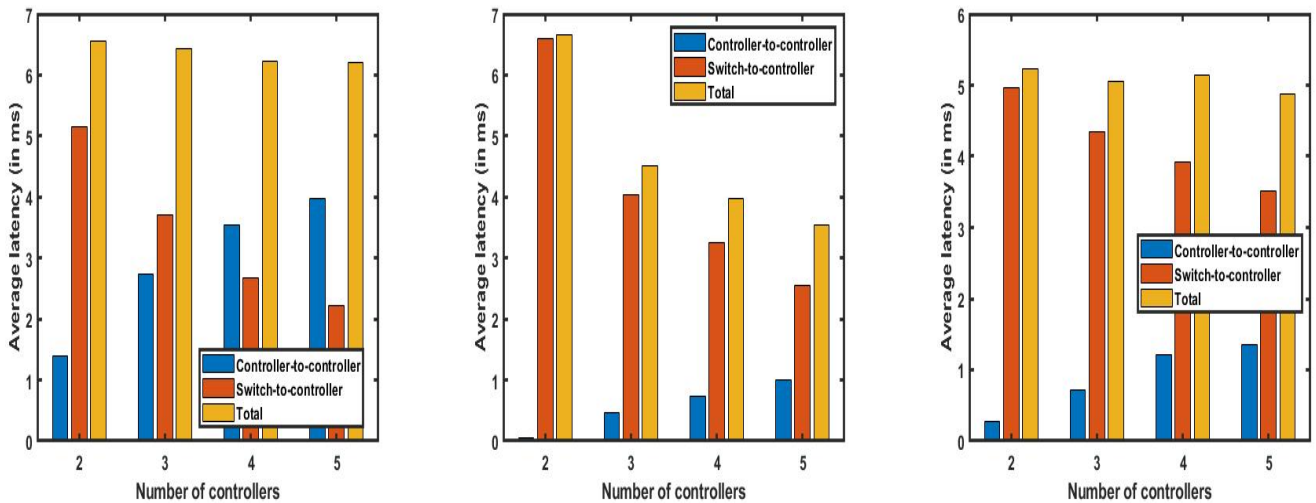

Fig. 2. Average Case latency (a) NSFNET (b) Agis and (c) Geant

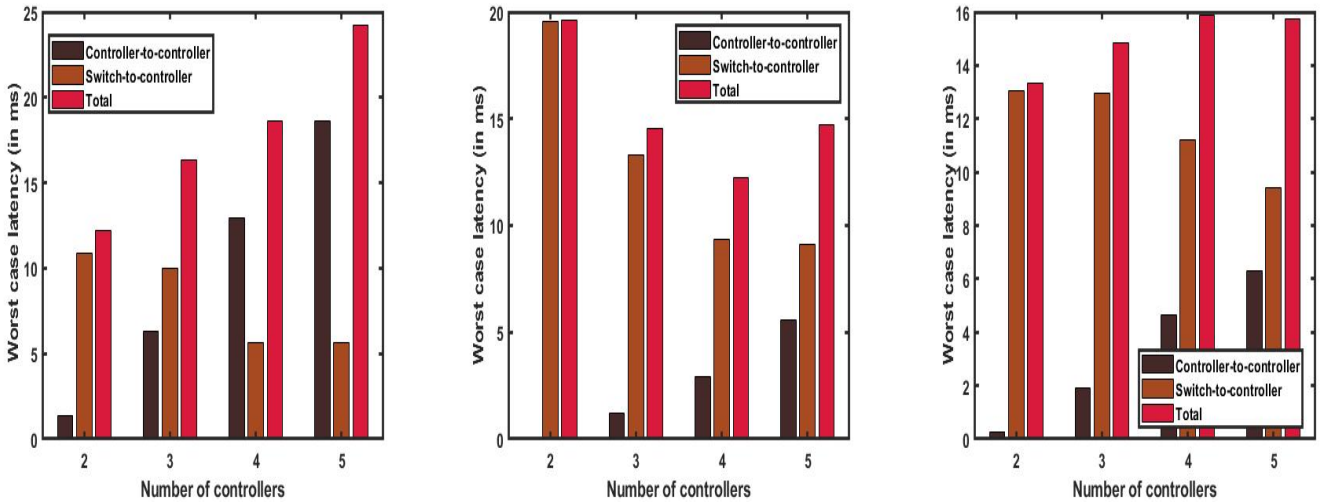

Fig. 3. Worst case latency (a) NSFNET (b) Agis and (c) Geant

[14] Chai, R., Yuan, Q., Zhu, L. et al. Control plane delay minimization-based capacitated controller placement algorithm for SDN. J Wireless Com Network 2019, 282 (2019). https://doi.org/10.1186/s13638-019-1607-x

[15] K. A. Rasol Rasol and J. Domingo-Pascual, "Joint Placement Latency Optimization of the Control Plane," 2020 International Symposium on Networks, Computers and Communications (ISNCC), 2020, pp. 1-6, doi:

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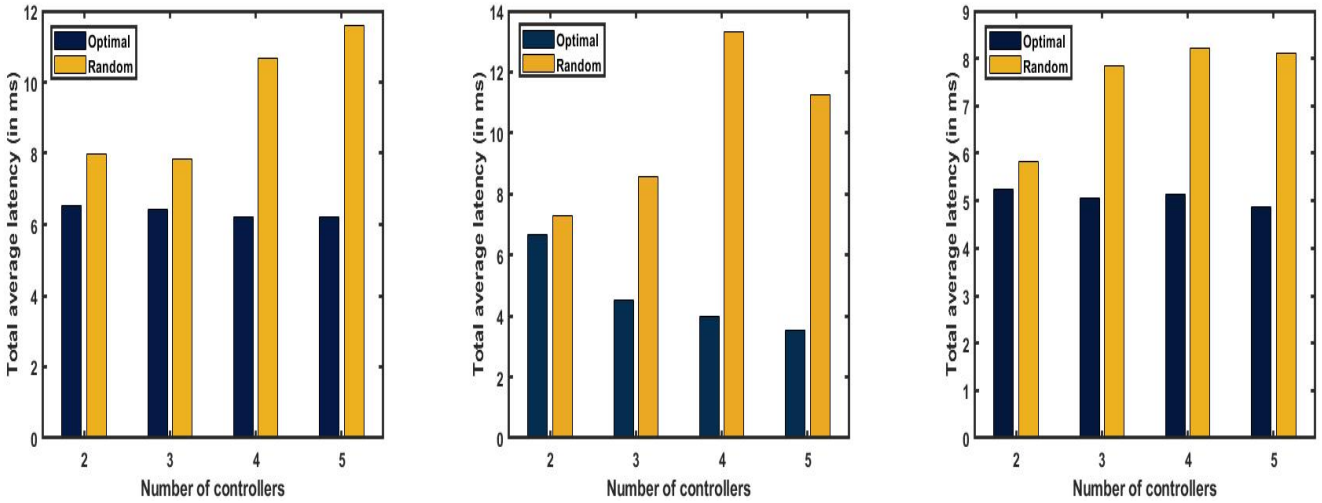

Fig. 4. Comparison with Random (Average Case latency) (a) NSFNET (b) Agis and (c) Geant

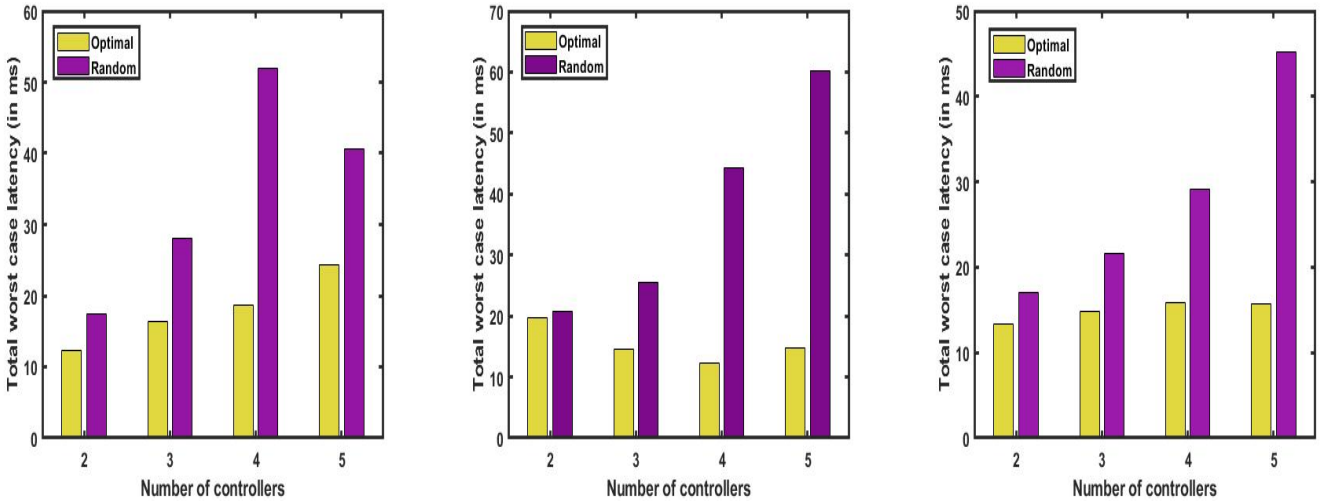

Fig. 5. Comparison with Random (Worst case latency) (a) NSFNET (b) Agis and (c) Geant

10.1109/ISNCC49221.2020.9297271.

[16] Alowa, Abdunasser, and Thomas Fevens. "Towards Minimum Inter-Controller Delay Time in Software Defined Networking." Procedia Computer Science 175 (2020): 395-402.

[17] A.K.Singh,S. Maurya and S. Srivastava, "Varna-based optimization: a novel method for capacitated controller placement problem in SDN," Frontiers of Computer Science, 14, 143402 (2020). https://doi.org/10.1007/s11704-018-7277-8

[18] http://www.topology-zoo.org/dataset.html.

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